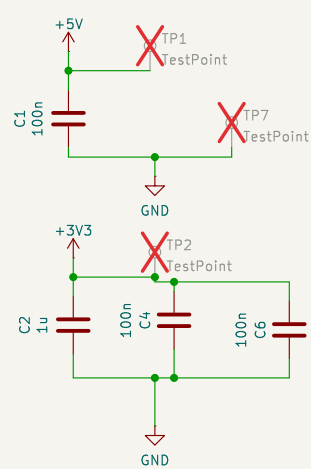
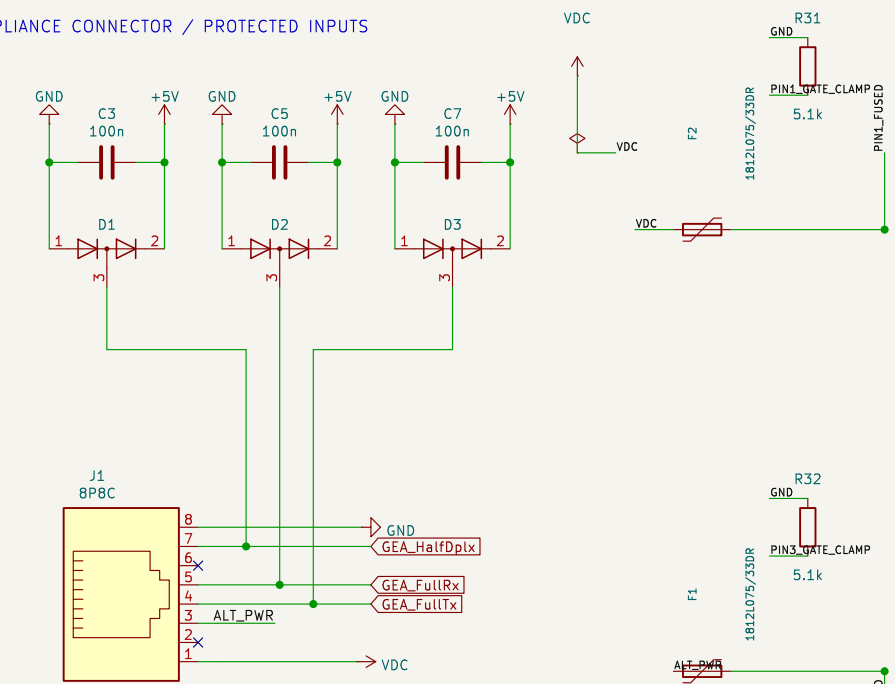
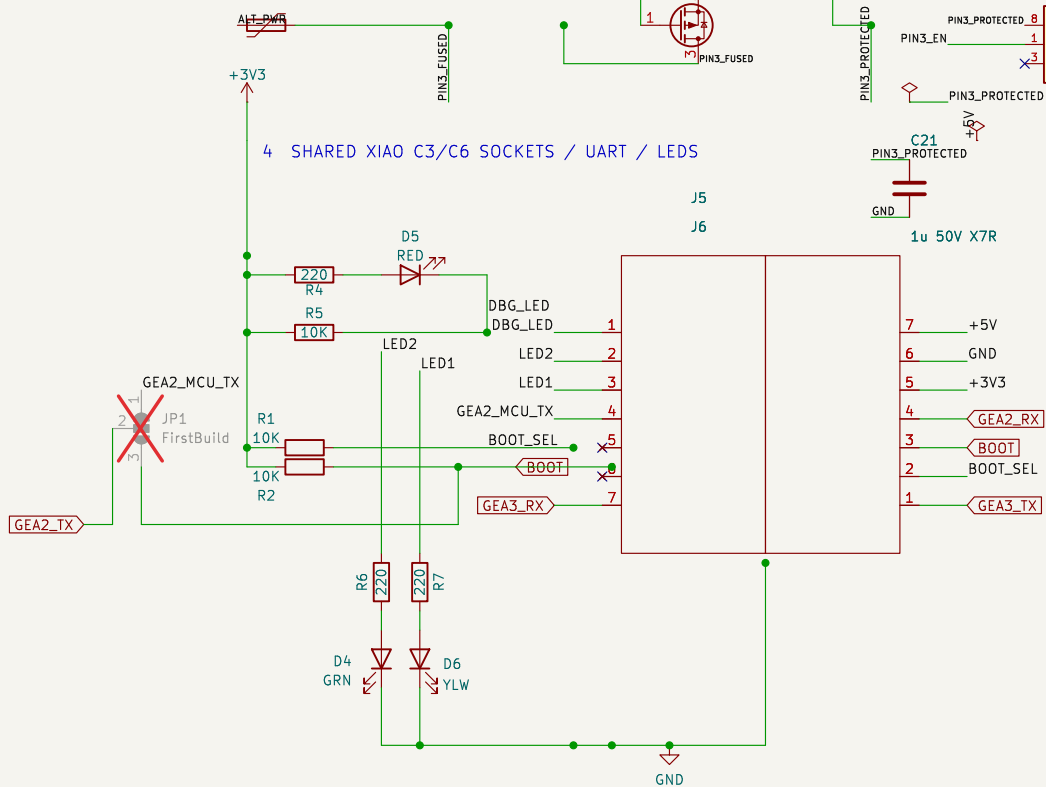


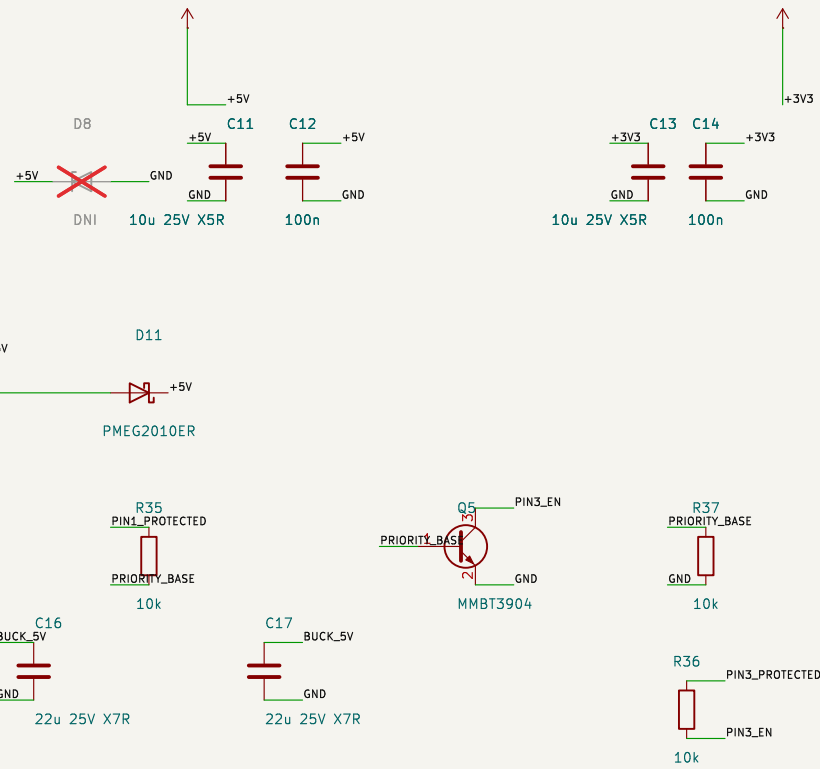
1 APPLIANCE CONNECTOR / PROTECTED INPUTS



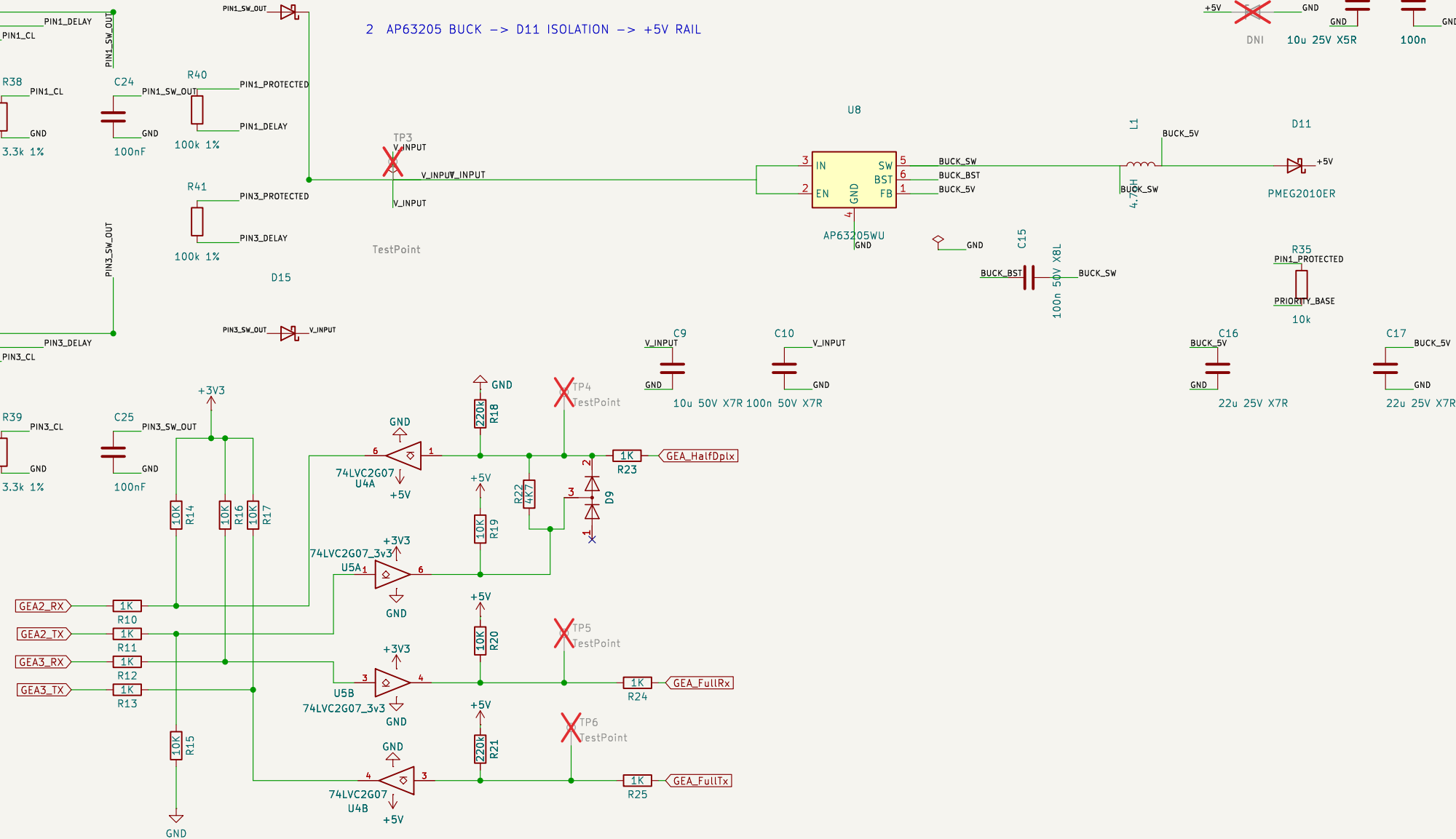
4 SHARED XIAO C3/C6 SOCKETS / UART / LEDS



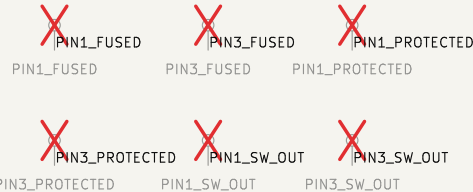
3 XIAO 3V3 OUTPUT / CARRIER LOGIC RAIL



2 AP63205 BUCK -> D11 ISOLATION -> +5V RAIL



5 INPUT TEST POINTS

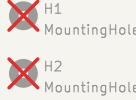


7 PROTECTED-INPUT TVS CLAMP



6 MODULE USB RECOVERY / SHARED-PIN NOTES

Use the installed module's own BOOT and RESET buttons. No carrier recovery header.  
D9 / BOOT is C3 GPIO9, but C6 GPIO20 (not C6 BOOT). C6 BOOT is not on the side headers.  
J5/J6 accept a pre-headered XIAO ESP32-C3 or C6; 3V3 supplies carrier logic only. No battery/underside pads are connected. Native USB remains on the module.  
R1/R2/R5 retain C3 strap pull-ups. JP1 default 1-2 uses D3; engineering alternate 2-3 uses D9.  
Disconnect the appliance cable before connecting module USB. Choose matching C3/C6 firmware.



J5/J6 are two separate 1x07 female sockets for a user-installed XIAO ESP32-C3 module.  
Native USB remains on the module; RESET is via the onboard button. No factory-module underside pads are connected.

POWER AND OPERATING NOTES

Appliance pin 1 / pin 3 selection is automatic (U9/U10 load switches, D14/D15 blocking); D1/D2/D3 are input ESD clamps.  
U8 AP63205 buck output BUCK\_5V reaches the +5V rail through D11, which blocks backfeed into the buck.  
J5/J6 accept a pre-headered XIAO ESP32-C3 or C6; 3V3 supplies carrier logic only. No battery/underside pads are connected. Native USB remains on the module.  
Both modules tie header 5V to USB VBUS. D11 protects the buck, not the USB host; no simultaneous-source protection is provided.  
CONNECT ONLY ONE SUPPLY AT A TIME; USB or appliance. Physically disconnect the appliance cable before USB connection.  
U9/U10 TPS1H200: FAULT NC, DIAG\_EN and exposed pad GND; DELAY via 100k to own VS. CL 3.3k = provisional 0.606 A. Retry 35-45 ms on / 0.8-1.2 s off; bench qualification required. DNP = do not populate; test points and mounting holes are not assembled components.

UNQUALIFIED PROTOTYPE - REVIEW BEFORE FABRICATION

Dr-Crow

Sheet: /  
File: GEA-Adapter-Rev3C.kicad\_sch

Title: GE Appliance Adapter - Rev3C shared XIAO C3/C6 carrier

Size: A2 Date: 2026-10-02

Rev: 3C

KiCad E.D.A. 9.0.9

Id: 1/1